PREP 2 · LESSONS 3–4

Drop tests and ramps

Repeats and means · bar charts · choosing a scale · plotting a line graph

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define accuracy. DN 851 [1]

How close a measurement is to the true value.

2 What is the independent variable? DN 870 [1]

The one variable that is deliberately changed.

3 Name the type of graph used when the independent variable is continuous. DN 882 [1]

A line graph

Key ideas LESSONS 3–4

Show your working

4 A ball is dropped onto three materials. The table shows its bounce height.

Material	Trial 1 / cm	Trial 2 / cm	Trial 3 / cm	Mean / cm
Foam	12	14	13	13
Bubble wrap	20	23	20	21
Carpet	31	28	31	30

a Calculate the mean bounce height for bubble wrap. Write it in the table. [1]

(20 + 23 + 20) ÷ 3 = 63 ÷ 3 = 21 cm ✓

b Which material is the best shock absorber? Give the reason. [1]

Foam, because it gives the lowest (mean) bounce ✓

c Explain why these results are shown on a bar chart. [1]

The independent variable (material) is categoric ✓

5 In the ramp practical, a trolley leaves a ramp and travels across the floor.

Ramp height / cm	5	10	15	20	25
Mean distance / cm	22	41	63	80	101

a Plot the last two points on the graph. (20, 80) ✓ (25, 101) ✓ [2]

b On a different graph, the largest value on the y-axis is 118 cm and there are 10 big squares. Draw a ring around the best value for each big square: [1]

10 cm	15 cm	(20 cm)	50 cm

c Name one control variable in the ramp practical. [1]

e.g. same trolley, same floor surface ✓

